

COMMUNITY ENGAGEMENT PROJECT (CEP)

SKILL BRIDGE

Community Employment and Skill Matching Portal

Submitted in Partial Fulfilment

of the Requirements for the Award of the Degree

of

BACHELOR OF SCIENCE

[To be added: Information Technology / Data Science]

SEMESTER V

By

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Under the esteemed guidance of

[To be added: Guide name]

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Department of Computing

VSIT | Vidyalankar School of
Information Technology

RE-ACCREDITED : GRADE 'A' BY NAAC

DEPARTMENT OF COMPUTING

VIDYALANKAR SCHOOL OF INFORMATION TECHNOLOGY

(Autonomous College affiliated to University of Mumbai)

MUMBAI, 400037

MAHARASHTRA

2026 - 2027

VIDYALANKAR SCHOOL OF INFORMATION TECHNOLOGY

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CERTIFICATE

I hereby certify that Vidhi Mahato (Roll Number: 24302A0028), student of Vidyalankar School of Information Technology, studying in T.Y. B.Sc. [To be added: Information Technology / Data Science], Semester V, has completed a C.E. project titled “Skill Bridge — Community Employment and Skill Matching Portal” at Khalsa College, Matunga, during the academic year 2026–27.

Internal Guide

Head of Department

Department of Computing

Internal Examiner

External Examiner

Date:



Principal

ORGANIZATION VISIT OBSERVATION SHEET

Attach the Organization Visit Observation Sheet duly signed by the reporting officer and mentor.

Organization: Khalsa College, Matunga

Student: Vidhi Mahato | Roll Number: 24302A0028

[To be added: Signed observation sheet covering all three visits]

PROJECT GUIDE MEETING RECORD

Attach the Project Guide Meeting Record duly signed by the mentor.

Project: Skill Bridge — Community Employment and Skill Matching Portal

Student: Vidhi Mahato | Roll Number: 24302A0028

[To be added: Signed guide meeting record]

DECLARATION

I, Vidhi Mahato, studying in T.Y. B.Sc. [To be added: Information Technology / Data Science], Semester V, at Vidyalankar School of Information Technology, submit this report on the Community Engagement Project “Skill Bridge — Community Employment and Skill Matching Portal” for the academic year 2026–27.

The fieldwork consisted of three visits to Khalsa College, Matunga. Our team collected real student responses by manually circulating a Google Form. The technical descriptions in this report concern the shared application developed for the project.

Sources used in preparing the report are identified in the references. Supporting forms and signatures are to be attached in their prescribed locations.

Signature of the Student with Date: _____

Vidhi Mahato

Roll Number: 24302A0028

ACKNOWLEDGEMENT

I am grateful to Vidyalankar School of Information Technology and the University of Mumbai for including community engagement in our academic work. The project allowed us to connect technical development with a student community.

My thanks go to our mentor, [To be added: Guide name], and to Khalsa College, Matunga, where the three visits took place. I also acknowledge the students who contributed their responses through the Google Form.

Working with Dhiraj Tendulkar, Gaurav Mhatre and Varun Patil made this a shared learning experience. I thank them for their part in the common project and recognise the contribution of everyone who supported its completion.

ABSTRACT

This Community Engagement Project presents Skill Bridge, a web portal intended to make placement support easier to follow for students and coordinators. The fieldwork was carried out through three visits to Khalsa College, Matunga. We collected genuine responses by manually sharing a Google Form among students. The work focuses on the connection between understanding student needs and providing a system through which opportunities, applications, training and guidance can be managed.

Students use a personal workspace to maintain profile information, explore openings, read matching explanations, apply for opportunities and enrol in workshops. They can also submit feedback and request guidance. Coordinators receive separate management views for reviewing applications and handling operational work. Administrators maintain protected configuration and recovery functions. The access model is enforced through server checks rather than interface visibility alone.

The portal uses React and TypeScript, supported by Vite, Tailwind CSS and Recharts. Express provides the API, and SQLite stores related records. A rule-based scoring process compares department, role preference, skills, location and a detected academic cutoff. The output is advisory: it explains relevance without claiming to determine employer selection.

Technical validation included compilation, isolated workflows, database lifecycle checks and production serving. The latest recorded run passed 157 automated checks, along with TypeScript and the production build. Lint completed with warnings. The results show that the tested functions operate as expected, but direct user acceptance and live-host verification remain separate activities. Skill Bridge offers a structured support mechanism aligned with learning and employment preparation. Confirmed field-response statistics, signed feedback and evidence of organisational adoption are to be added from the original records.

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CHAPTER 1: INTRODUCTION

1.1 Project Background

Our CEP work brought the team to Khalsa College in Matunga on three occasions. Students were the primary community for the project, and we distributed a Google Form to collect their responses manually. Skill Bridge was developed around placement support: finding suitable opportunities, understanding skill requirements and following up on applications. This report presents the work from my perspective while retaining the common facts and outcomes of the team project.

Table 1: Project team

Member	Roll number
Dhiraj Tendulkar	24302A0004
Vidhi Mahato	24302A0028
Gaurav Mhatre	24302A0067
Varun Patil	24302D0074

1.2 Problem Statement

A placement-support system must do more than display vacancies. Students need to understand relevance, eligibility and the next action, while coordinators need visibility into applications and training participation. The project therefore addresses the lack of a connected workflow between opportunity discovery, skill preparation and follow-up. The statement describes the problem our solution targets; a quantitative baseline for the college is [To be added].

1.3 Need for the Project

A common portal can reduce the effort required to move between opportunity information, applications and support requests. It also makes responsibilities visible: students manage their own records, coordinators handle operational decisions, and administrators maintain the system. This separation is important for both usability and data protection. Any measured reduction in administrative effort remains to be established through a live evaluation.

1.4 Objectives

We translated the project purpose into observable system functions. The primary objectives describe what users should accomplish; the secondary objectives concern the quality and continuity of those functions.

Primary Objectives

- Provide a distinct student workspace and coordinator workspace with server-enforced access.
- Support discovery, saving and application for opportunities with visible status updates.
- Connect workshop participation, feedback and guidance with coordinator follow-up.

Secondary Objectives

- Explain opportunity matching through understandable rules and reasons.
- Preserve related records and reject invalid or unauthorized changes.
- Provide aggregate research reporting while protecting individual responses.
- Maintain repeatable tests and practical operating instructions.

1.5 Scope

This report covers the developed portal and the associated three-visit CEP work. The system manages placement-support records and presents research information through role-appropriate views. It does not promise employment, automate employer selection or establish a measured improvement in placement rates. Production-mode operation has been tested locally; a live hosting address and evidence of organisational handover are [To be added].

1.6 SDG Alignment

Skill Bridge connects most directly with SDG 4 and SDG 8. Training discovery and guidance support learning beyond the classroom, while opportunity information and application tracking support preparation for work. The contribution is at the level of access and coordination. A numerical assessment of SDG outcomes would require follow-up evidence after actual use.

CHAPTER 2: REQUIREMENT ANALYSIS AND LITERATURE REVIEW

2.1 Methodology Adopted

The project used a field-based approach for gathering needs and an iterative approach for software development. All three visits were at Khalsa College, Matunga. We circulated a Google Form manually to obtain student responses, then related the project requirements to the portal's modules. Collection dates and the validated response count are [To be added]. Technical verification followed implementation and did not substitute for evaluation by actual users.

Table 2: Fieldwork record

Item	Confirmed information or record
Organization	Khalsa College, Matunga
Visits	Three visits, all at the same college
Collection method	Google Form manually circulated among students
Visit dates	[To be added]
Confirmed response count	[To be added]
Google Form link and response export	[To be added]
Verified findings and response summary	[To be added]

2.2 Stakeholder Interactions

The confirmed interaction involved students receiving the Google Form during the team's work at Khalsa College. The four members shared the same fieldwork setting and project. Coordinator and administrator needs are represented in the software design. Specific reporting-officer details, mentor-approved interaction notes and signed consultation records are [To be added].

2.3 Domain Overview

Placement support involves several connected records: student profiles, opportunities, applications, training sessions and requests for assistance. Skill matching helps a student interpret relevance, but formal eligibility and selection still depend on the stated recruitment criteria. The project uses this domain structure to connect information, participation and staff decisions.

2.4 Existing Process Analysis

For this project, students supplied information through a manually distributed Google Form. A form is suitable for collecting structured answers, but a placement portal must also maintain changing records after collection. Applications, enrolments and guidance requests therefore need additional workflows. Verified details of the college’s pre-existing operational process are [To be added].

2.5 Existing Solutions and Related Work

Our documented comparison is between a survey collection tool and the proposed placement-support application. The Google Form captures answers at submission; Skill Bridge maintains continuing activities such as application status and workshop participation. A literature review should also include verified published work where available. Research-paper and book references are [To be added].

2.6 Technologies Studied

The technical study concentrated on the tools used to build the application. React organises screens into components, TypeScript checks data contracts, and Express provides the server endpoints. SQLite supplies relational storage without a separate database service. Vite, Tailwind CSS and Recharts complete the build, presentation and charting requirements.

Table 3: Technologies used

Technology	Role in Skill Bridge
React	Component-based frontend and role-specific pages
TypeScript	Typed models and compile-time checks for frontend and server
Vite	Frontend development server and production bundling
Tailwind CSS	Interface layout and responsive styling

Technology	Role in Skill Bridge
Recharts	Research charts and visual summaries
Node.js and Express	Runtime, API routes, authentication and authorization
SQLite through node:sqlite	Persistent relational records, foreign keys and transactions
tsx and oxlint	TypeScript execution and lint checks
Google Forms	Manual circulation and collection of real student field responses

2.7 Research Gap and Need for Proposed Solution

The proposed solution is needed because response collection and ongoing service delivery are separate tasks. The portal brings opportunity discovery, applications, workshops and guidance into a consistent workflow. Its contribution is this integrated implementation for the project context, rather than a new machine-learning method or an experimentally proven research breakthrough.

2.8 Challenges Identified

Development required careful handling of permission boundaries, malformed content and database initialisation. Another challenge was ensuring that a successful interface message reflected an actual server save. Collection-related difficulties and their frequency are [To be added] from the field records. The final solution was reviewed against these technical concerns through isolated tests.

CHAPTER 3: PROPOSED SOLUTION

3.1 Solution Overview

The solution is a web portal that gives each role the tools needed for its work. Students manage their own placement activities, coordinators operate shared queues, and administrators handle system maintenance. A common API and database connect these views so that a recorded decision can be retrieved in the appropriate workspace.

3.2 System Architecture

The architecture has three main layers: the React user interface, the Express application server and SQLite storage. The interface does not grant authority by itself. Each protected request is checked on the server before database access. Configuration controls the port, database location and allowed origins; the production server can also deliver the compiled frontend.

[To be added]

Figure 1: System Architecture of Skill Bridge (placeholder)

3.3 Process Flow Diagram

The process moves from sign-in to a role-specific workspace. Students explore options and submit actions; the server validates them and records them. Staff then review the relevant queue and respond. The updated state is retrieved by the student. Failed or unauthorized requests should produce a clear error rather than a success message.

[To be added]

Figure 2: Process Flow of Skill Bridge (placeholder)

3.4 Module Description

Each module represents a recognisable task rather than an isolated page. Profiles support matching, opportunities support applications, and workshops support preparation. Feedback and guidance provide follow-up, while reporting and administration serve different access needs.

Table 4: Functional modules

Module	Responsibility
Authentication and access	Sessions, login/logout, role checks and account isolation
Student profile and matching	Profile preferences, skills and explained opportunity relevance
Opportunities and applications	Listing discovery, saved openings, submission and review status
Workshops	Session management, capacity, enrolment and attendance
Feedback and guidance	Student requests and coordinator responses
Research	Restricted response records and public aggregate reporting
Study and visit records	Three-visit documentation, action plans and evidence fields
Administration	Protected content, approved diagnostics and backup/restore

3.5 Advantages of Proposed Solution

The portal offers a consistent place to find opportunities, act on them and follow the response. Students receive matching reasons instead of an unexplained score, and staff use dedicated management views. Database-backed records support continuity across sessions. Long-term benefits require observation after actual deployment.

CHAPTER 4: SYSTEM DESIGN

4.1 System Design Overview

System design begins with the records needed by each workflow. Users own personal actions, opportunities receive applications, and workshops receive enrolments. TypeScript describes the expected structures, and SQLite enforces relational rules. Server-side validation provides another boundary before data is stored.

4.2 Use Case Diagram

Actor-based design makes access expectations explicit. Visitors can explore permitted public information, students use personal placement tools, coordinators manage operational records, and administrators maintain protected settings. This distinction is reflected in both navigation and API authorization.

Table 5: Actors and access

Actor	Permitted scope
Public visitor	Permitted public pages, opportunity information and aggregate research
Student	Own profile, saved listings, applications, enrolments, feedback and guidance
Placement Coordinator	Operational management, decisions, rosters and restricted research
Administrator	Protected configuration, diagnostics, backups and authorized management

[To be added]

Figure 3: Use Case Diagram of Skill Bridge (placeholder)

4.3 Workflow Diagram

The detailed workflow includes validation and failure paths. A repeated application or enrolment should not create another record. Workshop capacity must be checked during the operation, and a rejected request must leave the prior state intact. Successful staff updates become visible through subsequent data retrieval.

[To be added]

Figure 4: Application and Workshop Workflow of Skill Bridge (placeholder)

4.4 Module Design

Internally, the pages use a common context and API service to retrieve and update data. Express endpoints apply the relevant checks and database operations. Opportunity matching uses fixed rules with visible reasons, allowing students to understand why a listing is considered relevant. The score should not be interpreted as a likelihood of employment.

Database records and relationships

Table 6: Database entities

Entity or group	Purpose
users; sessions	Account identities, roles and active authentication sessions
student_profiles	Student-specific skills, interests and academic information
opportunities; applications	Listings linked to student submissions and review decisions
saved_opportunities	Student-specific saved listings
training_sessions; workshop_enrollments	Workshop information and participation relationships
feedback_submissions; guidance_requests	Requests and staff follow-up
survey_responses	Structured responses with provenance and controlled access

Entity or group	Purpose
employer_outreach	Staff-managed employer contact records
system_documents; app_metadata	Content documents and database initialisation state

[To be added]

Figure 5: Key Components of the Skill Bridge Database (placeholder)

Opportunity matching process

Table 7: Matching score components

Component	Points	Implemented rule
Department	40	Awarded when the listing includes the profile department
Preferred role	25	Case-insensitive containment between a preferred role and listing title
Skill overlap	Up to 20	Seven points per overlapping listed skill, capped at 20
Location	10	Preferred location matches, or the listing uses remote work
Academic cutoff	5	Entered percentage meets a detected two-digit percentage cutoff

The score is the sum of the awarded components, bounded between 0 and 100. Skill points are calculated as $\min(20, 7 \times \text{number of matching listing skills})$. A score of 75 or more is labelled High Fit; 50–74 is Moderate Fit; 30–49 is Partial Fit; lower scores are General Opportunity.

Text comparisons use case-insensitive containment, so the result depends on the terms entered in the profile and listing. The eligibility check detects a two-digit percentage cutoff and returns an eligibility caution where appropriate. It is not a complete eligibility parser. Students must review the original criteria even when the fit score is high.

CHAPTER 5: PROJECT IMPLEMENTATION

5.1 Development Methodology

The application was developed through repeated implementation and validation. Features were reviewed in relation to the intended student and coordinator workflows, then specific faults were corrected. The final checks included compilation, isolated API workflows and production-mode serving. Existing project data was kept separate from test data.

5.2 Module-wise Implementation

Each functional area is implemented through a page, shared service calls and server endpoints. Authentication establishes the current account; context exposes the relevant state to components. The server determines whether an operation is permitted and writes accepted changes to SQLite. This connects the proposed solution directly to the source organisation.

Table 8: Implementation mapping

Source area	Implementation role
src/pages	User-facing pages
src/components	Reusable interface components
src/context/AppContext.tsx	Shared session and application state
src/services/apiService.ts	Client API communication
src/utils/matching.ts	Rule-based matching function
server/index.ts	API and frontend-serving entry point
server/db.ts	Schema and database initialisation
server/validation.ts	Input and restoration validators
server/config.ts	Environment and runtime configuration
scripts	Tests, provisioning and backup utilities
README.md; .env.example	Setup instructions and configuration examples

Authentication and data integrity

Passwords are hashed with scrypt and a salt. Session tokens are carried in an HttpOnly cookie, and protected routes verify the session and role. Public registration is restricted to the student role. Database uniqueness constraints help prevent duplicate applications and enrolments. Restore operations validate supported content and relationships before transactional updates.

Research summaries

The reporting process groups responses by the requested filters and computes counts, distributions and rating summaries. Non-rated responses such as “Have not attended” are not treated as numeric satisfaction scores. Public reporting uses aggregate endpoints; individual responses are limited to authorized staff. The actual field-response totals and numerical findings are [To be added].

5.3 User Interface Solution

The user interface is organised around the task and role. Personal student records are presented separately from coordinator queues and administrative controls. Feedback messages reflect API outcomes, and input checks help prevent invalid submissions. Layout styling supports different screen sizes; final device-based usability observations are [To be added].

5.4 Screenshots of Developed Solution

Screenshots should document the implemented screens rather than a proposed mock-up. The reserved figures below identify the required views; use the current application and redact personal details before insertion.

(add picture)

Figure 6: Student Workspace of Skill Bridge (placeholder)

(add picture)

Figure 7: Placement Coordinator Workspace of Skill Bridge (placeholder)

(add picture)

Figure 8: Research Summary View of Skill Bridge (placeholder)

CHAPTER 6: TESTING AND VALIDATION

6.1 Test Cases

The test approach included positive and negative cases. It checked whether permitted actions succeeded and whether duplicate, malformed or unauthorized requests were rejected. Isolated databases made these checks repeatable. The table summarises the principal scenarios rather than listing every assertion.

Table 9: Representative test cases

Area	Scenario	Expected result
Authentication	Invalid credentials and unauthenticated requests	Rejected
Role authorization	Student requests staff-only resources	Rejected
Account isolation	One student requests another student's records	Private records not exposed
Application lifecycle	Submit, reject duplicate, record coordinator decision	Expected records and status retained
Workshop lifecycle	Enrol, reject duplicate, cancel, enforce capacity	Participation rules enforced
Research privacy	Public/student access to raw responses	Blocked; aggregate reporting available
Content and restoration	Malformed documents and invalid relationships	Rejected; valid restoration supported
Diagnostics	Client-supplied SQL and approved query identifiers	Raw SQL blocked; approved queries work
Production lifecycle	Fresh database, restart and admin provisioning	Startup and provisioning checks passed

6.2 Test Results

The available technical evidence shows a successful build and type check, followed by passing results in all four suites. The total is 157 reported checks. Lint warnings remain for review. The result is encouraging for the tested scope, while live hosting and complete visual interaction checks require their own evidence.

Table 10: Recorded automated results

Suite	Checks	Result
Acceptance	37	Passed
Workflow and security	97	Passed
Production lifecycle	14	Passed
Production serving	9	Passed
Total	157	Passed

6.3 User Acceptance Testing

A practical UAT session should cover finding a role, submitting an application, enrolling in training and reviewing a request from the coordinator side. Verified participant observations and approval are [To be added]. Passing automated checks does not replace this direct assessment of usability.

6.4 Feedback from Stakeholders

Stakeholder feedback should be presented from the actual responses and signed forms. The verified summary and any permission to quote participants are [To be added]. Annexure D provides an unfilled reporting-officer form so that feedback can be recorded without inventing an endorsement.

6.5 Performance Evaluation

The available results concern correctness and startup rather than measured speed. A production build and passing serving checks show that the application can run locally in production mode. Response-time and load measurements are [To be added]. They should be collected under a documented workload and hosting configuration.

CHAPTER 7: RESULTS AND IMPACT ASSESSMENT

7.1 Deliverables Submitted

The project produces a working web portal, its database implementation, test scripts and supporting instructions. This report and the annexures document those outputs. A signed receipt or approved handover record is [To be added], so availability of the artefacts is not presented as proof of institutional deployment.

7.2 Results Achieved

The application now connects the main placement-support functions through persistent records and differentiated access. The final recorded checks passed after the identified defects were corrected. Fresh production operation was also tested locally. Verified analysis of the real field responses remains [To be added] and is not inferred from test data.

7.3 Impact on Organization

The expected organisational benefit is better continuity between a student request and a staff response. Shared records can support follow-up and make responsibilities clearer. These benefits remain subject to adoption and evaluation at the college. No quantified reduction in effort or improvement in placement rates is claimed.

7.4 SDG Contribution

The project's SDG contribution lies in the services it makes available: learning support, career guidance and organised access to opportunities. These functions relate to SDG 4 and SDG 8. A stronger impact assessment would follow students over time and document actual use rather than rely on the presence of features alone.

CHAPTER 8: CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

The project brought together fieldwork at Khalsa College and the development of a role-based portal. Students supplied genuine responses through the circulated Google Form, while the software work produced connected placement and support functions. The recorded checks show technical progress. The remaining documentary evidence concerns field totals, photographs, approvals and direct user evaluation.

8.2 Key Findings

A key finding from development is that the visible interface depends on less visible safeguards. Role authorization, transactional restoration and safe initialisation protect the work performed through the screens. Another finding is the need to distinguish automated acceptance from stakeholder approval. Neither should be used as a substitute for the other.

8.3 Recommendations

The next practical steps are controlled deployment, user acceptance and completion of the evidence record. Staff should agree who maintains listings, responds to requests and manages backups. Public reports should remain aggregate, with individual responses restricted. Field photographs and signed forms should be attached before submission.

8.4 Future Enhancements

Future development could add carefully controlled notifications, more configurable matching and reporting across longer periods. Any extension should preserve privacy and be evaluated with users. A documented accessibility review and host-level performance study would complement the existing technical tests.

REFERENCES

Research Papers

[To be added: Verified research papers actually consulted, formatted in MLA style.]

Books Referred

[To be added: Books actually consulted, with verified MLA bibliographic details.]

Website

[To be added: Websites actually consulted and the topics referred to, with verified URLs and access details.]

Project source material:

Skill Bridge Project Team. Skill Bridge: Community Employment and Skill Matching Portal. Project source code, 2026.

Skill Bridge Project Team. "Project Setup and Verification." README.md, 2026.

Skill Bridge Project Team. "Opportunity Matching." src/utis/matching.ts, 2026.

Skill Bridge Project Team. "Survey Questionnaire." src/data/defaultSurveySeed.ts, 2026.

Skill Bridge Project Team. "Verification Suites." scripts/verify-acceptance.ts, scripts/test-workflows.ts, scripts/test-production-startup.ts, and scripts/verify-production-serving.ts, 2026.

ANNEXURES

The annexures retain the labels prescribed in the supplied format: A, B, C, D, G and H.

Annexure A: Field Visit Photographs

Visit 1 — Khalsa College, Matunga

Date of visit: [To be added]

(add picture)

Figure 9: Visit 1 at Khalsa College, Matunga

Visit 2 — Khalsa College, Matunga

Date of visit: [To be added]

(add picture)

Figure 10: Visit 2 at Khalsa College, Matunga

Visit 3 — Khalsa College, Matunga

Date of visit: [To be added]

(add picture)

Figure 11: Visit 3 at Khalsa College, Matunga

ANNEXURE B: SURVEY QUESTIONNAIRE

Our field collection used a Google Form shared manually among students. This annexure records the questionnaire implemented in the project. The original form URL and confirmation of its final wording are [To be added].

1. Which department or course are you studying in?

Response options: Computer Science (CS); Information Technology (IT); Data Science (DS).

2. What is your current year of study?

Response options: First; Second; Third; Fourth; Other.

3. What job role or career domain are you mainly interested in?

Response options: Open text response.

4. How many campus recruitment drives offered roles relevant to your preferred domain during the current academic year?

Response options: 0; 1–2; 3 or more; Not sure.

5. Is the placement cell currently meeting your placement-related needs?

Response options: Yes; Partly; No; Not enough experience to judge.

6. How often do you receive placement information early enough to prepare and apply?

Response options: Always; Sometimes; Rarely; Never; Not applicable.

7. How useful has the placement training you attended been for your preferred role?

Response options: 1–5, where 1 is Not useful and 5 is Very useful; Have not attended.

8. What is the biggest barrier you face in getting a suitable placement?

Response options: Limited relevant roles; Eligibility restrictions; Skill gaps; Late information; Interview preparation; Location or pay mismatch; Other; No major barrier.

9. Which placement support do you need most urgently?

Response options: Relevant employer connections; Technical training; Aptitude preparation; Resume and interview help; Career guidance; Timely alerts; Internship support; Other.

10. How satisfied are you with overall placement support?

Response options: 1–5, where 1 is Very dissatisfied and 5 is Very satisfied; Not enough experience to judge.

11. Briefly describe your experience with the placement process.

Response options: Open text response.

12. What one change would most improve placement support for you?

Response options: Open text response.

ANNEXURE C: INTERVIEW QUESTIONS

The project includes an interview guide for placement-coordination discussions. The questions below reproduce that guide. Interview dates, respondent identity, consent and recorded answers are [To be added]. Illustrative answers in the software are not reproduced as actual staff statements.

1. How are student preferences and skills collected?
2. How many relevant employers and roles were available by department?
3. What explains differences in opportunities?
4. How are new employers approached?
5. What makes specialist employers difficult to attract?
6. How are notices received and published?
7. How are training needs identified?
8. How is training effectiveness measured?
9. What alternatives exist for students excluded by eligibility criteria?
10. Are duties, location, and pay details checked before publication?
11. What staffing, time, or budget constraints exist?
12. Which concerns are already being addressed?
13. What additional actions are feasible, with owners and timelines?
14. What can Skill Bridge support?
15. How is student feedback reviewed and answered?

ANNEXURE D: REPORTING OFFICER FEEDBACK FORM

Blank form for assessment of the prepared project

Project: Skill Bridge — Community Employment and Skill Matching Portal

Organization: Khalsa College, Matunga

Student: Vidhi Mahato | Roll Number: 24302A0028

Reporting officer name: _____

Designation: _____

Date of review: _____

Relevance of the project to student/community needs

Usefulness and clarity of the developed solution

Observations on the demonstration

Suggested improvements

Overall assessment and acceptance status

Reporting officer signature: _____

Organization stamp, if required: _____

ANNEXURE G: SOURCE CODE REPOSITORY

Repository link: [To be added]

Submitted version or commit identifier: [To be added]

Contents uploaded and verified by: [To be added]

The following inventory identifies the project contents to include in the submission. It does not confirm that a repository has already been published.

Table 11: Repository contents

Source area	Contents
src/pages	User-facing pages
src/components	Reusable interface components
src/context/AppContext.tsx	Shared session and application state
src/services/apiService.ts	Client API communication
src/utils/matching.ts	Rule-based matching function
server/index.ts	API and frontend-serving entry point
server/db.ts	Schema and database initialisation
server/validation.ts	Input and restoration validators
server/config.ts	Environment and runtime configuration
scripts	Tests, provisioning and backup utilities
README.md; .env.example	Setup instructions and configuration examples

Exclude passwords, private environment files, real student records and local database backups from any public repository. Provide authorised access to sensitive evidence separately.

ANNEXURE H: USER MANUAL

The instructions below describe how to use the completed workflows. Access depends on the signed-in account. The deployment address and support contact are [To be added].

Student workflow

1. Start by signing in to the student account and checking the accuracy of profile details.
2. Browse the opportunities and use filters to narrow the list.
3. Read the fit explanation as an aid, then verify the stated eligibility and deadline.
4. Apply to a selected opening or save it; return to the application record for updates.
5. Enrol in an appropriate workshop if capacity is available.
6. Use feedback or guidance for additional support, and log out after the session.

Coordinator workflow

1. Open the coordinator workspace using the assigned account.
2. Create or update opportunity and training information after checking its accuracy.
3. Work through the application queue and save the appropriate decision.
4. Review enrolments and attendance from the workshop roster.
5. Handle guidance and feedback requests through their management screens.
6. Check the returned status after each update and sign out when the work is complete.

Administrator workflow

1. Sign in through the provisioned administrative account.
2. Review configuration and use the registered diagnostics for maintenance.
3. Create a backup before performing sensitive changes.
4. Use the restore control with an appropriate backup file and confirm the result.
5. Protect the backup files and avoid sharing administrative credentials.

Local setup and production preparation

Install a Node.js version compatible with the project dependencies and native SQLite support, then run npm install in the project folder.

Use `npm run dev` for local development. The configured API and frontend addresses should be read from the terminal output.

Run `npm run typecheck`, `npm run build` and `npm test` before preparing a release.

For production, set `NODE_ENV=production`, a persistent `DB_PATH` and `CLIENT_URL` matching the actual website origin. Configure HTTPS and the intended reverse proxy.

Provision an administrator with explicit credentials using `npm run admin:provision -- <username> <password>`. Use protected credential handling in the deployment environment.

Run `npm start` after the build. Verify login, authorised changes and direct route refreshes on the actual host.

Use `npm run backup:db` for the database backup utility. Test recovery with a disposable copy before relying on a backup. Do not run demo reset commands against the real database.

If an operation fails

Check whether the action was saved before trying again. Verify the account role and server connection, then use the reported message to guide the next step. Escalate persistent errors with a clear description and no confidential records.